Cervais New Guardian Phase 2 Deliverable Tracking

Advanced Security & Intelligence Layer (9 weeks)

Project Timeline

Project Start Date:
February 2, 2026

Project Duration:
9 weeks (2 months)

Deliverables

Scope	Deliverable	Description	Milestone Date	Target	Notes
Sprint 1: Networking / Hardware/ Architecture					
Multi-Transport Circle of Trust (CoT) Connectivity & Auto-Detection		Design and implement a transport-agnostic Circle of Trust (CoT) capability for SG-X Guardian devices that enables secure peer-to-peer trust establishment and communications across heterogeneous network types, including wired LAN (Ethernet), Bluetooth, Wi-Fi, cellular (LTE/5G), and satellite links. CoT membership, Trust establishment, and policy enforcement shall be independent of the underlying network transport and automatically adapt to available interfaces without requiring manual configuration.	Milestone 1	2026-02-02	
Hardware Security	Secure Element Integration	Integrate embedded secure element chip (dedicated security co-processor) to establish hardware root of trust. Implement secure element driver, initialize cryptographic subsystem, and establish protected key storage. The secure element provides tamper-resistant key storage and cryptographic operations isolated from main processor.	Milestone 1	2026-	Foundation for hardware attestation
Hardware Security	Hardware Key Manager	Migrate from software-based key storage to secure element-backed Device Key Pair (DKP). Generate and store cryptographic keys within secure element's protected memory. Keys never exposed to software layer - all signing operations performed within hardware boundary. Implement key lifecycle management (generation, rotation, revocation).	Milestone 1	2026-	Secure key storage
Hardware Security	Hardware Attestation Protocol	Implement secure element-based remote attestation protocol. Device generates cryptographic proof of its hardware and firmware integrity state. Implement challenge-response attestation flow where verifier sends nonce, device produces signed attestation quote containing platform measurements. Verifier validates quote signature and measurements against known-good baselines.	Milestone 1	2026-	Hardware trust verification
Nebula Mesh	Nebula CA Operations	Initial installation and implementation of Nebula mesh networking framework. Install Nebula binaries on Guardian devices, configure Nebula daemon for startup. Implement Certificate Authority operations for Circle-based mesh networking. Circle owner generates CA key pair, signs member certificates binding DID to overlay IP address. Implement certificate issuance workflow: member requests cert, CA validates Circle membership via Verifiable Credential, issues time-limited cert with groups and permissions. Support cert renewal and key rotation. This deliverable establishes the foundational Nebula infrastructure for all subsequent mesh networking capabilities.	Milestone 1	2026-	Mesh cert management

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Nebula Mesh	Overlay Network Assignment	Configure Nebula's virtual network interface creation and IP assignment. Implement virtual IP address management for Nebula overlay network (192.168.100.0/24 per Circle). Circle owner assigns unique overlay IP to each member Guardian. Track IP allocations in CircleOfTrust entity (next_overlay_ip counter). Configure Nebula to create virtual network interface (nebula0) with assigned IP on each Guardian. All peer-to-peer traffic flows through encrypted overlay tunnels using these virtual IPs. Includes initial Nebula configuration file generation and network interface initialization.	Milestone 1	2026-	Virtual networking
Nebula Mesh	Lighthouse Configuration	Install and configure Nebula lighthouse nodes to enable NAT traversal and peer discovery. Deploy lighthouse infrastructure on publicly-accessible Guardians or cloud instances. Lighthouses maintain registry of peer locations (public IP:port mappings). Configure peers to register with lighthouse on Nebula daemon startup, update on network changes. When peer A wants to connect to peer B, lighthouse provides B's current public endpoint for UDP hole punching. Support multiple redundant lighthouses per Circle for availability. Includes lighthouse selection criteria, deployment procedures, and initial connectivity testing.	Milestone 1	2026-	Mesh connectivity
Nebula Mesh	Multi-Hop Relay Routing	Configure and implement Nebula's multi-hop relay routing capabilities for Guardians that cannot establish direct UDP connections (symmetric NAT, restrictive firewalls). Enable relay functionality in Nebula configuration. When direct connection fails, route traffic through intermediate relay peers. Discover relay paths using Nebula's distributed routing protocol. Each relay Guardian forwards encrypted packets (double-encrypted: original + relay layer). Track relay bandwidth usage and enforce configurable relay limits through Nebula configuration parameters. Enables connectivity for satellite/cellular Guardians with challenging network conditions. Includes relay performance testing and optimization.	Milestone 1	2026-	Network resilience
Sprint 2: Hardware Security Enhancement					
Hardware Security	PCR Measurement	Implement Platform Configuration Register (PCR) measurement collection during boot sequence. PCRs are secure element registers that accumulate cryptographic hashes of firmware, bootloader, kernel, and configuration. Extend PCR values at each boot stage (PCR0=BIOS, PCR4=bootloader, integrity "fingerprint" for device state. Store golden reference PCRs for known-good configuration.	Milestone 2	2026-	Integrity baseline

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Hardware Security	Secure Boot Chain	Implement secure boot chain with verified boot sequence. Each boot stage verifies cryptographic signature of next stage before execution. BIOS verifies application signature. PCR measurements recorded at each stage. If signature verification fails, boot halts. Prevents execution of tampered or malicious firmware. Establish chain of trust from hardware root (secure element) through entire software stack.	Milestone 2	2026-	Boot integrity
Demonstration	Hardware Attestation Demo	Live demonstration of 3-node Circle performing mutual hardware attestation. Show Guardian A challenging Guardian B's integrity, B generating signed attestation quote containing current PCR values, A verifying quote signature and comparing PCRs against golden baseline. Demonstrate detection of tampered device (modified firmware) - attestation fails when PCR values don't match expected baseline. Showcase automatic quarantine of compromised Guardian.	Milestone 2	2026-	Milestone demo
Sprint 3: Identity & Trust Enhancement					
DID/VC	W3C DID Implementation	Implement W3C Decentralized Identifiers (DID) Core 1.0 standard with custom did:guardian method. Each Guardian assigned persistent globally-unique DID (format: did:guardian:<base58-hash>). DID derived from device serial number and secure element public key. DIDs remain constant across device lifetime, firmware updates, and network changes. Implement DID method specification defining creation, resolution, update, and deactivation operations. DIDs replace ephemeral email-based identity for Circle membership.	Milestone 3	2026-	Persistent identity anchor
DID/VC	DID Document	Implement DID Document creation, publishing, and resolution. DID Document is JSON-LD structure containing public keys, authentication methods, and service endpoints for a DID. Guardian generates DID Document on first boot containing secure element public key, Nebula distributed storage (IPFS or Circle-local registry). Implement DID resolver to fetch and validate DID Documents. Support key rotation by updating DID Document without changing DID itself.	Milestone 3	2026-	Key discovery

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DID/VC	Verifiable Credentials	Implement W3C Verifiable Credentials (VC) for cryptographically-provable Circle membership. Circle owner issues signed VC to member's DID asserting membership status, assigned role (owner/member), join date, and permissions. VC contains issuer DID (Circle owner), subject DID (member), claims (role, permissions), issuance/expiration dates, and cryptographic proof (owner's signature). Members present VCs during peer authentication to prove Circle membership without contacting central authority. Support VC revocation via status list.	Milestone 3	2026-	Decentralized membership
DID/VC	DID Resolution Service	Implement DID resolution service enabling Guardians to discover peer public keys and endpoints from DIDs. Resolver accepts DID as input, queries distributed storage or local DID registry, retrieves and validates DID Document, returns public key and service endpoints. Resolution handles caching (1hr TTL), cache invalidation on key rotation, and fallback to multiple resolution sources for reliability. Essential for establishing peer connections - Guardian only needs peer's DID to initiate secure channel.	Milestone 3	2026-	DID lookup
DID/VC	VirtualID-DID Integration	Update VirtualID computation to incorporate base DID, establishing two-layer identity architecture. New formula: VirtualID = SHA256(DID || Current_DKP_PubKey || PCR_values || policy_digest || Nonce_I || Nonce_R). DID provides persistent identity anchor while VirtualID provides session-scoped unlinkable credential. VirtualID changes when PCR values or policy changes, forcing re-attestation. Prevents correlation across sessions while maintaining DID-based long-term trust. This hybrid approach combines standards compliance (DID) with novel privacy properties (VirtualID unlinkability).	Milestone 3	2026-	Two-layer identity
Sprint 4: Distributed Coordination (CRL Gossip)					
CRL Gossip	CRL Data Structure	Design Certificate Revocation List (CRL) data structure for peer-to-peerrevocation propagation. CRL entry contains: revoked DID, device_id, user_id, Circle_id, revocation reason (compromised/lost/stolen/policy violation),severity level, timestamp, revoker's DID, cryptographic signature. CRL storedin distributed CertificateRevocationList entity. Each entry cryptographically signed by issuer (Circle owner or member reporting compromise).	Milestone 4	2026	

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CRL Gossip	Gossip Protocol	Implement epidemic-style gossip protocol for decentralized CRL propagation without central authority. Each Guardian maintains local CRL copy. Periodically (every 1-5 minutes), Guardian randomly selects peer and exchanges CRL updates. Peer merges received revocations into local CRL, then forwards to other peers. Uses probabilistic flooding with anti-entropy mechanisms. Track which peers received each revocation in "peers_notified" list. Mark CRL entry as "propagated" after reaching threshold (e.g., 80% of Circle). Achieves eventual consistency across all Circle members even with network partitions.	Milestone 4	2026-	Distributed revocation
CRL Gossip	Emergency Revocation	Implement priority emergency broadcast channel for critical revocations (compromised devices, active attacks). When Guardian marked "severity: critical", immediately broadcast REVOCATION_NOTICE to all connected peers, bypassing normal gossip intervals. Receiving Guardians prioritize forwarding emergency revocations before routine gossip. Terminate all active sessions with revoked DID instantly. Send push notifications to users. Emergency revocations propagate to 90%+ of Circle within 30 seconds vs 5-10 minutes for normal gossip. Essential for containing active breaches.	Milestone 4	2026-	Critical security
CRL Gossip	Offline Revocation Sync	Implement offline revocation queue for Guardians operating in disconnected environments (tactical ops, remote locations, satellite with intermittent connectivity). When Guardian goes offline, queue outgoing revocations locally. Track pending revocations with retry counter and timestamps. When connectivity restored, synchronously push queued revocations to peers. Fetch missed revocations from peers by comparing CRL version vectors. Resolve conflicts using timestamp-based "last writer wins" or signature-based trust hierarchy. Ensures Guardians can revoke compromised peers even when isolated.	Milestone 4	2026-	Offline support
Demonstration	CRL Gossip Demo	Live demonstration of peer-to-peer revocation propagation in 5-node Circle. Scenario: Guardian C reported compromised, Circle owner issues emergency revocation for C's DID. Show revocation gossip spreading through network topology: C→A→B→D→E. Visualize propagation progress: which Guardians received revocation, propagation timestamps, gossip paths. Show revoked Guardian C automatically disconnected from all peers, marked red in UI. Demonstrate offline Guardian reconnecting later and syncing missed revocations. Verify revocation persists across Guardian restarts.	Milestone 4	2026-	Major milestone
Sprint 5: Intelligence Layer					

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Virtual Shift AI	Anomaly Detection Engine	Implement ML-based behavioral anomaly detection engine using lightweight edge-deployable models. Train baseline behavior profiles: normal traffic patterns, typical peer interaction frequencies, expected protocol sequences, device resource usage. Use streaming anomaly detection algorithms (Isolation Forest, One-Class SVM) to identify deviations. Detect anomalies: unusual traffic spikes, unexpected protocol violations, abnormal resource consumption, suspicious peer behavior. Generate anomaly scores with confidence levels. Trigger Virtual Shift policy updates when anomaly score exceeds threshold. Optimize for resource-constrained embedded hardware (10MB model size, <100ms inference time).	Milestone 5	2026-	AI threat detection
Virtual Shift AI	Automated Policy Adaptation	Implement automated Virtual Shift policy adaptation triggered by AI anomaly detection. When anomaly detected, AI engine generates policy update recommendations: tighten firewall rules, increase attestation frequency, enable additional logging, quarantine suspicious peers. Circle owner reviews AI recommendations and approves/rejects. Upon approval, Guardian signs and broadcasts VSHIFT_ALERT with new policy blob. All Circle members receive, verify signature, atomically apply policy update, rotate VirtualIDs (forcing re-attestation). Policy updates propagate via existing gossip infrastructure. Log all policy changes with AI justification for audit trail. Support manual override for false positives.	Milestone 5	2026-	Dynamic security
Virtual Shift AI	AI Network Optimization	Implement AI-driven network routing optimization adapting to real-time conditions. Monitor network metrics: peer latencies, packet loss rates, bandwidth utilization, relay hop counts. ML model predicts optimal routing paths based on current conditions and historical performance. Dynamically switch between direct P2P connections and multi-hop relay routes. Balance load across multiple relay paths for high-priority traffic. Predict network degradation (satellite weather interference, cellular congestion) and proactively reroute. Implement reinforcement learning for routing decisions - reward low-latency high-throughput paths, penalize congested routes. Reduce average latency 20-40% vs static routing.	Milestone 5	2026-	Performance optimization

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Virtual Shift AI	Threat Prediction Model	Implement predictive threat modeling for proactive security posture adjustment. Analyze temporal patterns in security events: time-of-day attack frequencies, attack type sequences, peer compromise cascades. ML model predicts likelihood of imminent attacks based on precursor indicators (reconnaissance scans, unusual traffic patterns, geographic threat intelligence). Generate threat forecasts: "High probability of DDoS attack in next 2 hours (85% confidence)". Preemptively strengthen security: increase attestation frequency, tighten firewall rules, enable intensive logging. Correlate predictions across Circle members - if Peer A compromised, predict Peer B at elevated risk (shared network segment). Reduce successful attack rates 30-50% through predictive hardening.	Milestone 5	2026-	Predictive security
Security Tools	Suricata IDS/IPS Integration	Integrate Suricata IDS/IPS engine for real-time network traffic analysis and threat detection. Install Suricata binaries on Guardian, configure rule sets (Emerging Threats, custom signatures). Implement packet capture integration with Guardian network interfaces. Parse Suricata EVE JSON logs, extract alerts (malware, exploits, policy violations). Feed Suricata alerts to AI anomaly detection engine for correlation with behavioral patterns. Support inline blocking mode - Suricata drops malicious packets before reaching applications. Configure signature auto-updates and rule management. Provides deep packet inspection complementing Guardian's AI-based detection.	Milestone 5	2026-	Cervais Security Team
Security Tools	NMAP Network Discovery	Integrate NMAP for automated network discovery and device profiling. Implement scheduled NMAP scans of Guardian's local network segment. Detect connected devices (IP, MAC, open ports, OS fingerprinting, service versions). Store discovered device inventory in ConnectedDevice entity. Cross-reference with approved device whitelist - flag unauthorized devices for approval. Integrate discovery results with AI threat prediction - identify vulnerable services (outdated SSH, unpatched web servers). Trigger vulnerability scans on newly discovered devices. Configure scan intensity (stealth/aggressive) and scheduling (hourly/daily). Essential for maintaining accurate network topology and identifying rogue devices.	Milestone 5	2026-	Cervais Security Team

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Demonstration	Virtual Shift Demo	Live end-to-end demonstration of Virtual Shift AI capabilities in 4-node Circle. Scenario: Inject simulated port scan attack targeting Guardian B. Show AI anomaly detection engine identifying unusual traffic pattern, calculating anomaly score (0.89/1.0). AI generates policy recommendation: "Block scanning source IP, increase attestation frequency from 5min to 1min". Circle owner receives alert, reviews AI justification, approves policy. Watch VSHIFT_ALERT propagate via gossip protocol. All Guardians apply new policy atomically, rotate VirtualIDs, perform immediate re-attestation. Demonstrate attack blocked after policy update. Show AI network optimization dynamically rerouting traffic around congested network path to maintain connectivity.	Milestone 5	2026-	Major milestone
Sprint 6: Observability & Optimization					
Forensics	Tamper-Evident Audit Trail	Implement tamper-evident audit trail using cryptographic Merkle tree chain-of-custody for forensic evidence integrity. Every security event (threat detections, policy changes, revocations, attestation failures) logged with timestamp, actor DID, event hash. Events organized into Merkle tree - each leaf is event hash, internal nodes are hash of children. Tree root hash signed with Guardian's secure element key and stored immutably. Any tampering (modified events, deleted logs, reordered timeline) invalidates Merkle proof. Implement append-only log storage with cryptographic linking - each log entry contains hash of previous entry. Supports court-admissible evidence with verifiable chain-of-custody. Export audit trail with Merkle proofs for external forensic analysis.	Milestone 6	2026-	Evidence integrity
Forensics	Evidence	Implement comprehensive forensic evidence collection during security incidents. Capture network forensics: full packet captures (pcap) of malicious traffic, connection metadata, DNS queries, TLS handshakes. System forensics: process lists, open file handles, memory dumps, system call traces. Application forensics: Guardian daemon logs, policy enforcement decisions, attestation results. Trigger automatic evidence collection on high-severity events (quarantine activation, revocation, attestation failure). Store evidence encrypted with secure element key in tamper-evident container. Implement evidence export API for security operations center (SOC) integration. Compress and deduplicate evidence to optimize storage (~100MB per incident). Evidence retained 90 days, then auto-archived.	Milestone	2026-	Incident

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Forensics	Attack Timeline Reconstruction	Implement intelligent attack timeline reconstruction through automated event correlation. Aggregate forensic evidence from multiple sources (network captures, system logs, AI anomaly alerts) into unified timeline. Correlate events across Circle members to trace attack propagation: initial compromise → lateral movement → data exfiltration. Use ML clustering to group related events. Identify attack kill chain phases: reconnaissance, exploitation, installation, command & control, exfiltration. Visualize attack flow with interactive timeline graph showing causal relationships between events. Highlight critical pivot points where intervention could have stopped attack. Export timeline reports (JSON, PDF) with annotated threat intelligence. Essential for incident response, root cause analysis, and security improvements.	Milestone 6	2026-	Investigation tools
Security Tools	OpenVAS Vulnerability Scanner	Integrate OpenVAS for comprehensive vulnerability assessment of Guardian-protected networks. Install OpenVAS scanner engine, configure vulnerability feeds (NVT updates, CVE database). Implement scheduled vulnerability scans: weekly full scans, daily quick scans for new devices. Scan targets include connected endpoints, IoT devices, network infrastructure discovered via NMAP. Parse OpenVAS XML reports, extract vulnerabilities (CVE IDs, severity scores, affected software). Prioritize critical vulnerabilities (CVSS 9.0+) for immediate alerting. Cross-reference vulnerabilities with active exploits (threat intelligence feeds). Generate remediation reports with patching recommendations. Integrate scan results with AI threat prediction - flag high-risk devices for enhanced monitoring.	Milestone 6	2026-	Cervais Security Team
Security Tools	ClamAV Malware Detection	Integrate ClamAV antivirus engine for malware detection and file scanning. Install ClamAV daemon (clamd) on Guardian, configure signature database auto-updates (hourly). Implement real-time file scanning: monitor file system events (inotify), scan newly created/modified files. Scan targets include USB-connected storage, network file shares, downloaded files, email attachments. Parse ClamAV scan results, extract malware signatures (trojan, ransomware, spyware). Automatically quarantine infected files - move to isolated directory, revoke permissions. Generate malware incident reports with file hash, signature name, detection timestamp. Integrate with forensic evidence collection - preserve quarantined samples for analysis. Feed malware detections to AI correlation engine for threat pattern analysis.	Milestone 6	2026-	Cervais Security Team

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Security Tools	Security Tool Orchestration	Implement unified orchestration layer for all security tools (Suricata, NMAP, OpenVAS, ClamAV). Create centralized alert aggregation pipeline - normalize alerts from different tools into common schema (timestamp, severity, source_tool, threat_type, affected_target). Implement correlation engine: cross-reference alerts across tools (NMAP discovers vulnerable SSH, OpenVAS confirms CVE, Suricata detects exploitation attempt, ClamAV finds malware). Generate unified security dashboard showing multi-tool threat visibility. Configure alert deduplication and priority ranking. Implement health monitoring for security tools (process status, signature update freshness, scan completion rates). Support tool configuration management via Guardian UI. Provides "single pane of glass" security for operations center functionality.	Milestone 6	2026-	Cervais Security Team
Performance	eBPF Enforcement	Replace user-space nftables firewall with kernel-level XDP/eBPF for line-rate packet filtering. eBPF programs execute in kernel at network driver level before packets reach networking stack - enabling 10Gbps+ filtering on commodity hardware. Compile VShift security policies into eBPF bytecode: firewall rules, rate limiting, protocol enforcement. Load eBPF programs into kernel using bpf() syscall. XDP performs early packet filtering - drop malicious packets at NIC before CPU processing. Implement eBPF maps for stateful tracking: connection tables, rate limit counters, blocked IP lists. Update policies dynamically without kernel recompilation. Reduces packet processing latency from 100µs (nftables) to <10µs (eBPF). Critical for high-throughput tactical networks.	Milestone 6	2026-	Performance optimization
Performance	Performance Benchmarking	Conduct comprehensive performance benchmarking of Phase 2 components under realistic workloads. Test secure element operations: key generation (1000 iterations), signing throughput (ops/sec), attestation latency (ms). Nebula mesh performance: throughput (Mbps) direct vs relay, handshake latency, concurrent connection scaling. CRL gossip propagation time vs Circle size (10/50/100 members). AI inference latency for anomaly detection. eBPF packet filtering throughput at 1Gbps/10Gbps line rates. Measure resource utilization: CPU, memory, storage I/O. Stress testing: 1000 simulated attacks/hour, policy updates under load, offline queue overflow scenarios. Generate performance report with graphs, bottleneck analysis, optimization recommendations. Establish performance baselines for regression testing.	Milestone 6	2026-	Optimization validation

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Documentation	Phase 2 Architecture Guide	Comprehensive architectural documentation for Phase 2 advanced security features (~150 pages). Sections: Executive Summary, System Architecture Overview, Hardware Security (secure element integration, PCR measurements, attestation protocol), Identity Management (DID/VC implementation, VirtualID integration), Distributed Coordination (CRL gossip, Nebula mesh networking), AI Intelligence Layer (anomaly detection, Virtual Shift adaptation), Forensics & Compliance (audit trails, evidence collection). Include architecture diagrams, component interaction flows, protocol specifications, API references, data schemas. Deployment architecture for various scenarios: on-premises, edge, tactical. Security analysis: threat model, attack surface, mitigation strategies. Performance benchmarks and optimization guidelines. Target audience: security architects, system integrators, DevSecOps engineers.	Milestone 6	2026-	Final deliverable
Documentation	Administrator Guide v1.0	Updated Administrator Guide covering Phase 2 operational procedures (~80 pages). New sections: DID lifecycle management (generation, rotation, recovery), secure element operations (key provisioning, attestation verification, PCR baselining), Circle-wide policy management, CRL emergency revocation procedures, AI anomaly threshold tuning, Virtual Shift policy review workflows. Troubleshooting guides: attestation failures, CRL gossip propagation issues, Nebula mesh connectivity problems, AI false positives. Operational runbooks: onboarding new Guardians with secure element, investigating security incidents using forensic tools, generating compliance reports. CLI command reference, configuration file formats, monitoring dashboards. Target audience: Circle owners, security administrators, SOC operators. Includes screenshots, configuration examples, decision trees for incident response.	Milestone 6	2026-	Final deliverable
Documentation	Phase 2 Test Report	Comprehensive test report documenting Phase 2 quality assurance (~60 pages). Test coverage summary: unit tests (secure element drivers, DID generation, VirtualID computation, CRL data structures), integration tests (end-to-end attestation flows, multi-node gossip propagation, Nebula mesh formation), performance tests (throughput benchmarks, latency measurements, scalability testing). Test results matrix: total tests executed, pass/fail rates, code coverage percentage, critical bugs fixed. Security testing: penetration test results, fuzzing coverage, cryptographic validation. Compliance validation: NIST control verification, CMMC requirement mapping. Known issues and mitigations. Acceptance criteria verification. Test environment specifications. Regression test suite for ongoing validation. Provides confidence in production readiness.	Milestone 6	2026-	Final deliverable

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Demonstration	Final Phase 2 Demo	End-to-end demo: secure element attestation, DID/VC, CRL gossip, Nebula mesh, AI, forensics	Milestone 6	2026-	Phase 2 completion

Important Notes

• Phase 2 assumes successful Phase 1 completion

• Hardware security features require Guardian devices with embedded secure element chip

• Cloud management portal is **optional** and not included in base Phase 2 scope

• DPDK integration deferred - eBPF provides sufficient performance improvement